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Computer Science & IT

Computer Networks

DPP: 1

MAC Layer

Q1 Consider a network using the pure ALOHA medium access control protocol, where each frame is of length 4KB. The channel transmission rate is 512 Kbps. Calculate vulnerable time (in milisecond)?

Q2 Consider a network using the slotted ALOHA medium access control protocol, where each frame is of length 5000 bits. The channel transmission rate is 1 Mbps. Calculate vulnerable time (in milisecond)?

Q3 Consider a network using the pure ALOHA medium access control protocol, where each frame is of length 1,000 bits. The channel transmission rate is 10 Kbps. The aggregate number of transmissions across all the nodes with a rate of 5 frames per second. Calculate throughput in percentage (rounded to the nearest integer)?

Q4 Consider a network using the slotted ALOHA medium access control protocol, where each frame is of length 5,00 bits. The channel transmission rate is 1 Kbps. The aggregate number of transmissions across all the nodes with a rate of 2 frames per second. Calculate throughput in percentage (rounded to the nearest integer)?

Q5 Consider a LAN with three nodes A, B, and C. Time is divided into fixed-size slots, and a node can begin its transmission only at the beginning of a slot. The probabilities of generation of a frame in a time slot by A, B, and C are 0.2, 0.3 and 0.4 respectively. The probability of sending a frame in the first slot without any collision by any of these three stations is _____.(upto 3 decimal places)

Q6 The minimum frame size required for a CSMA/CD based computer network running at 1 Mbps on a 2000 meter cable with a link speed of 2×10^6 meter per second is :

(A) 1000 bits (B) 1000 bytes
(C) 2000 bits (D) 2000 bytes

Q7 Determine the maximum length of the cable for transmitting data at a rate of 1 Mbps in an Ethernet LAN with frames of size 1000 bits. Assume the signal speed in the cable to be 20000 meter per second.

(A) 10 meter (B) 10 kilometer
(C) 20 meter (D) 20 kilometer

Q8 A and B are the only two stations on an Ethernet. Each has a steady queue of frames to send. Both A and B attempt to transmit a frame and collide. What is the probability that A wins the first backoff race?

(A) 0.25 (B) 0.5
(C) 0.75 (D) 1

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Answer Key

Q1	128	Q5	0.452
Q2	5	Q6	(C)
Q3	18	Q7	(A)
Q4	37	Q8	(A)

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Hints & Solutions

Q1 Text Solution:

Frame size = 4KB = $2^2 \times 2^{10} \times 2^3$ bits = 2^{15} bits

D.T.R = 512 kbps = $2^9 \times 10^3$ bits/sec

$t_x = \frac{\text{Frame size}}{\text{DTR}} = \frac{2^{15}\text{ bits}}{2^9 \times 10^3 \text{ bits / sec}} 2^6 \times 10^{-3}$ sec

t_x = 64 ms

Vulnerable time = 2 × t_x = 128 ms

Q2 Text Solution:

Frame size = 5000 bits = 5×10^3 bits

D.T.R = 1 Mbps = 10^6 bits/sec

$t_x = \frac{5 \times 10^3 \text{ bits}}{10^6 \text{ bits / sec}} = 5 \times 10^{-3}\text{sec} = 5\text{ ms}$

Vulnerable time = t_x = 5ms

Q3 Text Solution:

Frame size = 10^3 bits

D.T.R = 10 kbps = 10^4 bits/sec

$t_x = \frac{10^3 \text{ bits}}{10^4 \text{ bits / sec}} = \frac{1}{10}\text{sec}$

$1\text{sec} \rightarrow 5\,\text{Frames}$

$\frac{1}{10}\text{sec} \rightarrow \frac{5}{10}\text{Frames}$

$\left(t_x = \frac{1}{10}\text{sec}\right) \rightarrow \frac{1}{2}\ \text{Frames}$

$G = \frac{1}{2}$

G : Average number of transmission per frame time

S = G × e^{-2G}

$S = \frac{1}{2} \times G \times e^{-2G}$

$S = \frac{1}{2e}$

S = 0.1839 = 18.39%

$S \approx 18\%$

S : Average number of successful transmission per frame time.

Q4 Text Solution:

Frame size = 500 bits,

D.T.R = 1 kbps = 1000 bits/sec

$t_x = \frac{500 \text{ bits}}{1000 \text{ bits/sec}} = \frac{1}{2}\text{sec}$

$1\text{sec} \rightarrow 2\,\text{Frame}$

$t_x = \frac{1}{2}\text{sec} \rightarrow 1\,\text{Frame}$

S = G × e^{-G}

$S = \frac{1}{e}$

S = 0.3678 = 36.78%

$S \approx 37\%$

Q5 Text Solution:

P_1 = 0.2

P_2 = 0.3

P_3 = 0.4

Channel throughput = $P_1 \times (1 - P_2) \times (1 - P_3) + (1 - P_1) \times P_2 \times (1 - P_3) + (1 - P_1) \times (1 - P_2) \times P_3$

Channel throughput

= 0.2 × 0.7 × 0.6 + 0.8 × 0.3 × 0.6 + 0.8 × 0.7 × 0.4

= 0.084 + 0.144 + 0.224

= 0.452

Q6 Text Solution:

D.T.R = 1 Mbps = 10^6 bits/sec

Distance = 2000 m = 2×10^3m

Signal speed = 2×10^6 m/sec

$t_x \geq 2t_p$

$\frac{\text{Frame size}}{\text{D.T.R}} \geq 2 \times \frac{\text{Distance}}{\text{Signal speed}} \times \text{DTR}$

Frame size

$\geq 2 \times \frac{2 \times 10^3 \text{ m}}{2 \times 10^6 \text{ m/ sec}} \times 10^6 \text{ bits / sec}$

$\text{Frame size} \geq 2000\,\text{bits}$

Q7 Text Solution:

Frame size = 10^3 bits

D.T.R = 10^6 bits/sec

$t_x = \frac{10^3 \text{ bits}}{10^6 \text{ bits / sec}}$

$t_x = 10^{-3}$ sec

Ethernet CSMA/CD

Signal speed = 2×10^4 m/sec

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$t_x \geq 2t_p$

$10^{-3}\text{sec} \geq 2 \times \frac{\text{Distance}}{\text{Signal speed}}$

$10^{-3}\text{sec} \geq 2 \times \frac{\text{Distance}}{2 \times 10^4 \text{m/sec}}$

Distance $\leq 10^{-3}\text{sec} \times 10^4 \text{m/sec}$

$\text{Distance} \leq 10 \text{ meter}$

Q8 Text Solution:

Both A and B attempt to transmit a frame

Then collide so we can make the tabel.

A	B
0	0
0	1
1	0
1	1

P(A win) = 1/4 = 0.25

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